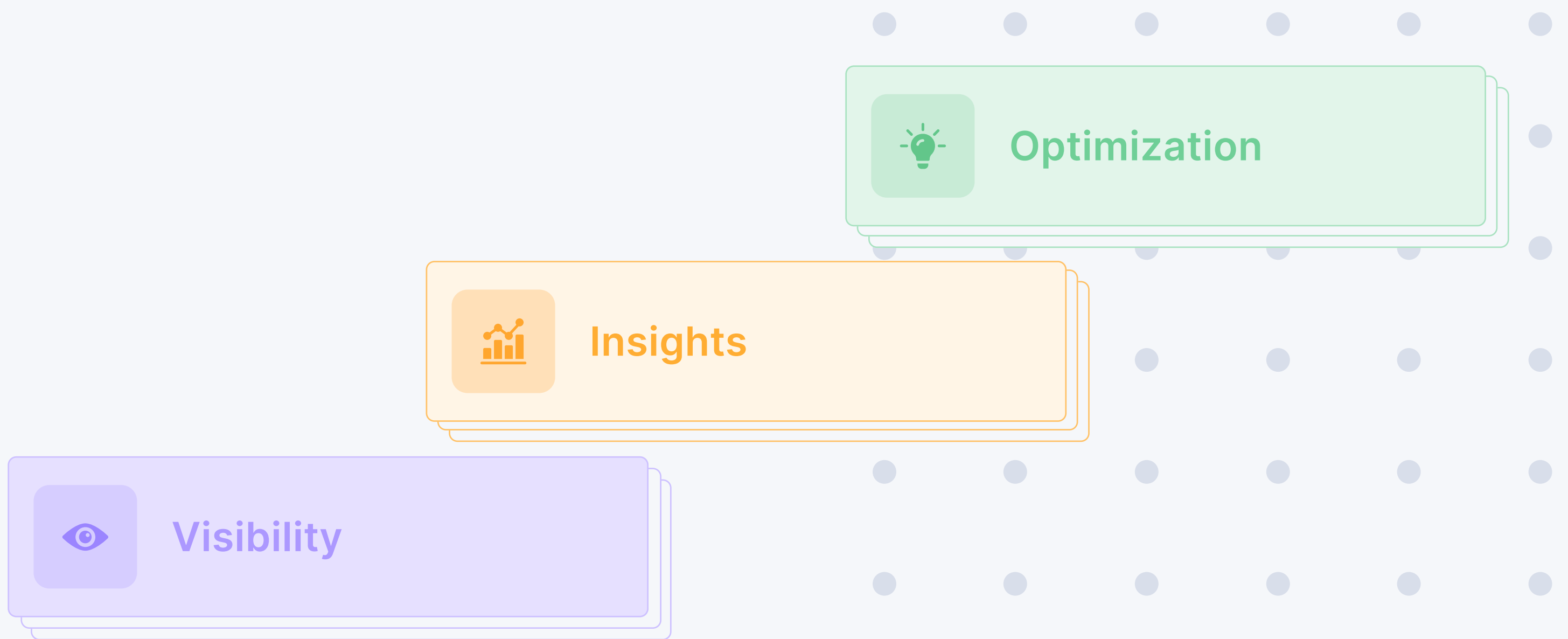




Engineering Leader's Guide to Resource Allocation

Operational excellence is key to engineering success and maintaining your organization's competitive advantage. Another critical aspect is understanding project resourcing, the associated costs, and using this telemetry to closely align engineering strategy to business priorities.



In this guide—part two of the five-part engineering success model—we’ll explore how to incorporate resource allocation into your **engineering metrics program** and show you how to refine your project investment strategy.

Here’s what we’ll cover in the pages ahead:

- **Visibility:** Getting a holistic picture of what everyone in the organization is working on
- **Insights:** Using industry standards and historical patterns to find reallocation opportunities
- **Optimization:** Fine tuning your project resourcing strategy to ensure engineering drives positive business outcomes

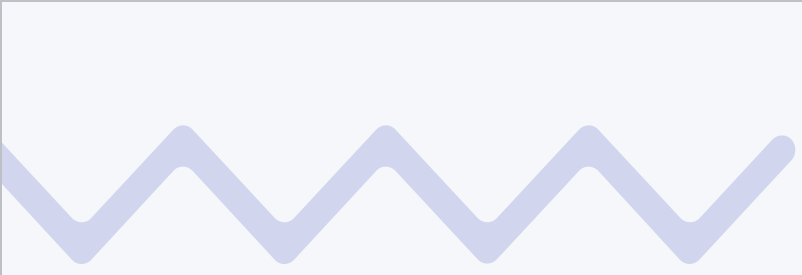
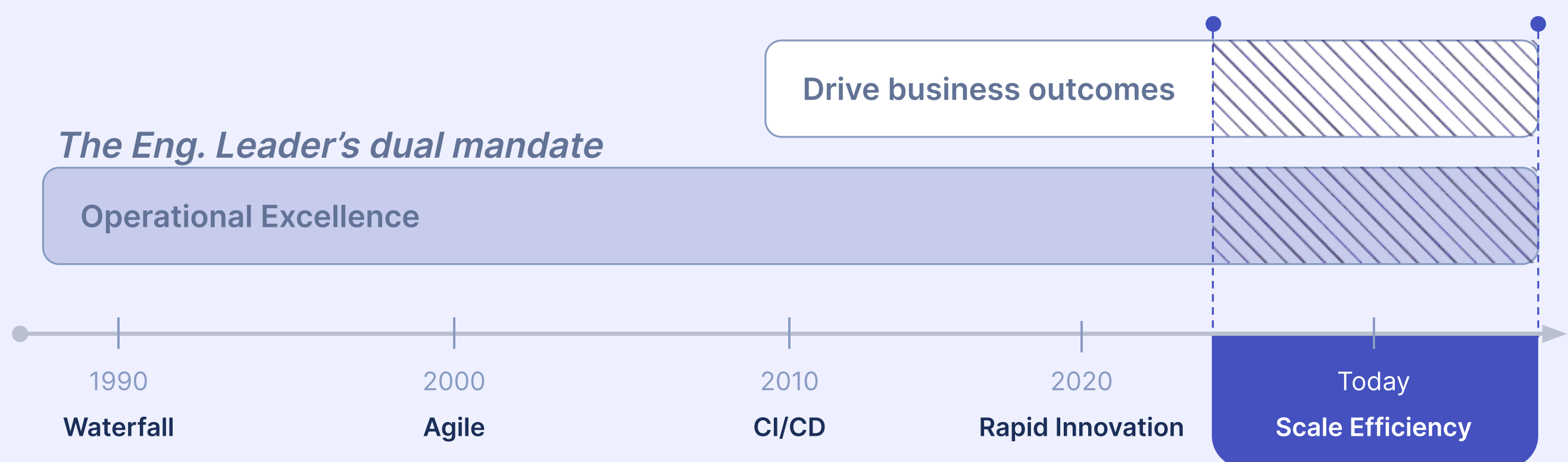


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Engineering Leaders Are Leading Businesses

Engineering leaders are responsible for ops and business outcomes



Too often for too long, engineering was seen as a cost center rather than a value driver. Thankfully, that perception is changing. Now it's nearly universally understood that every company is a software company, meaning engineering excellence needs to be a top concern for leadership teams, regardless of industry vertical.

It also means that engineering leaders are now responsible for BOTH delivering quality products as efficiently as possible AND positively impact the business.

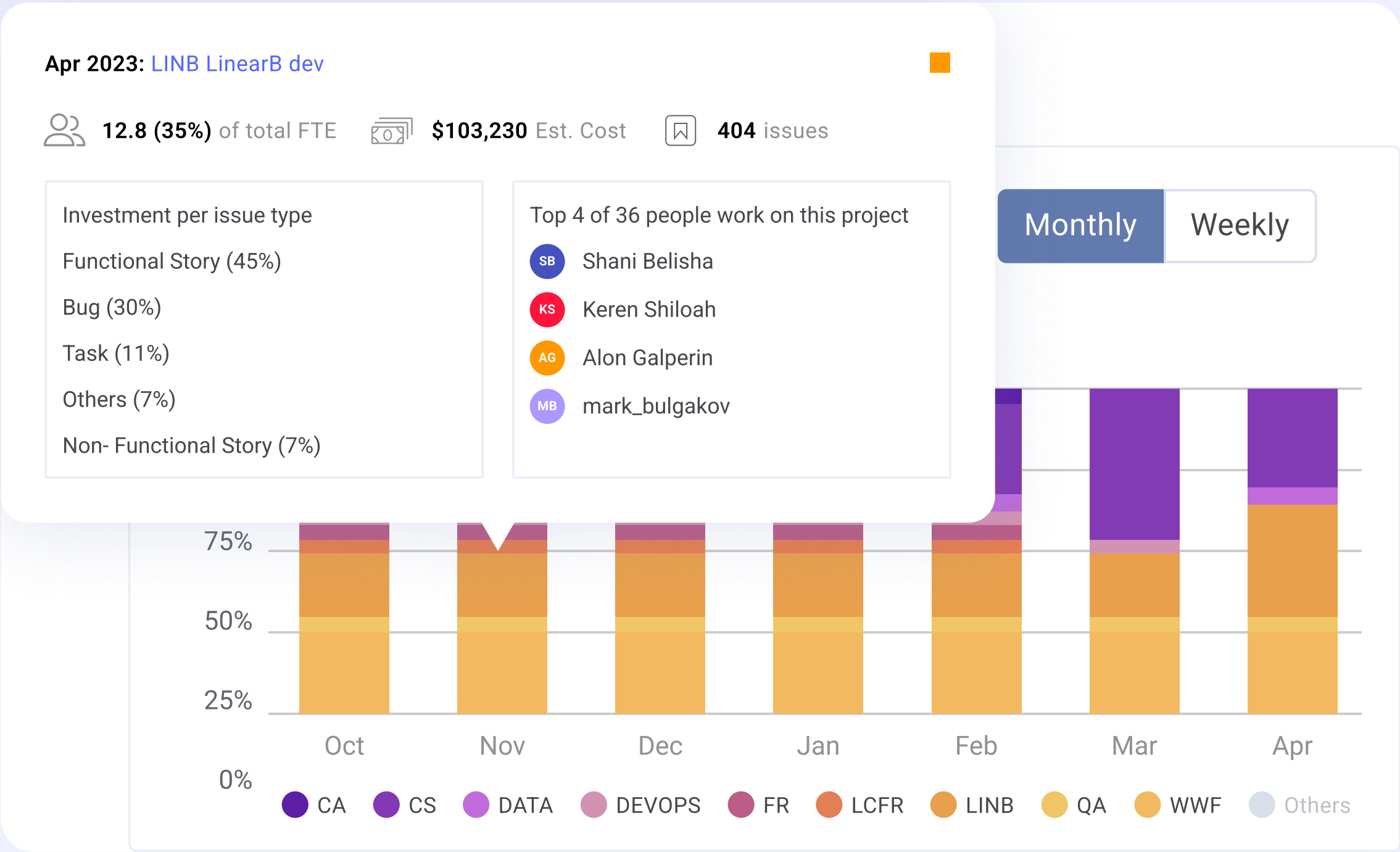
To drive business outcomes, work to identify and prioritize high value projects that business leaders expect to drive revenue, retention, and customer satisfaction.

To maintain operational excellence and complete those high value projects on time (further impacting the business), ensure you have enough of the right people assigned.

Both of those considerations require effective resource allocation as part of a holistic metrics program.

Three Outcomes of Resource Allocation Visibility

A holistic view of project resourcing and cost



We talked in a [previous guide](#) about **getting your metrics program off the ground** and the data that needs to be included to build it.

Understanding how projects/initiatives/work are resourced is a significant part of that metrics program. When leaders have visibility into this data, they'll see myriad benefits including:



Knowing what your teams are working on



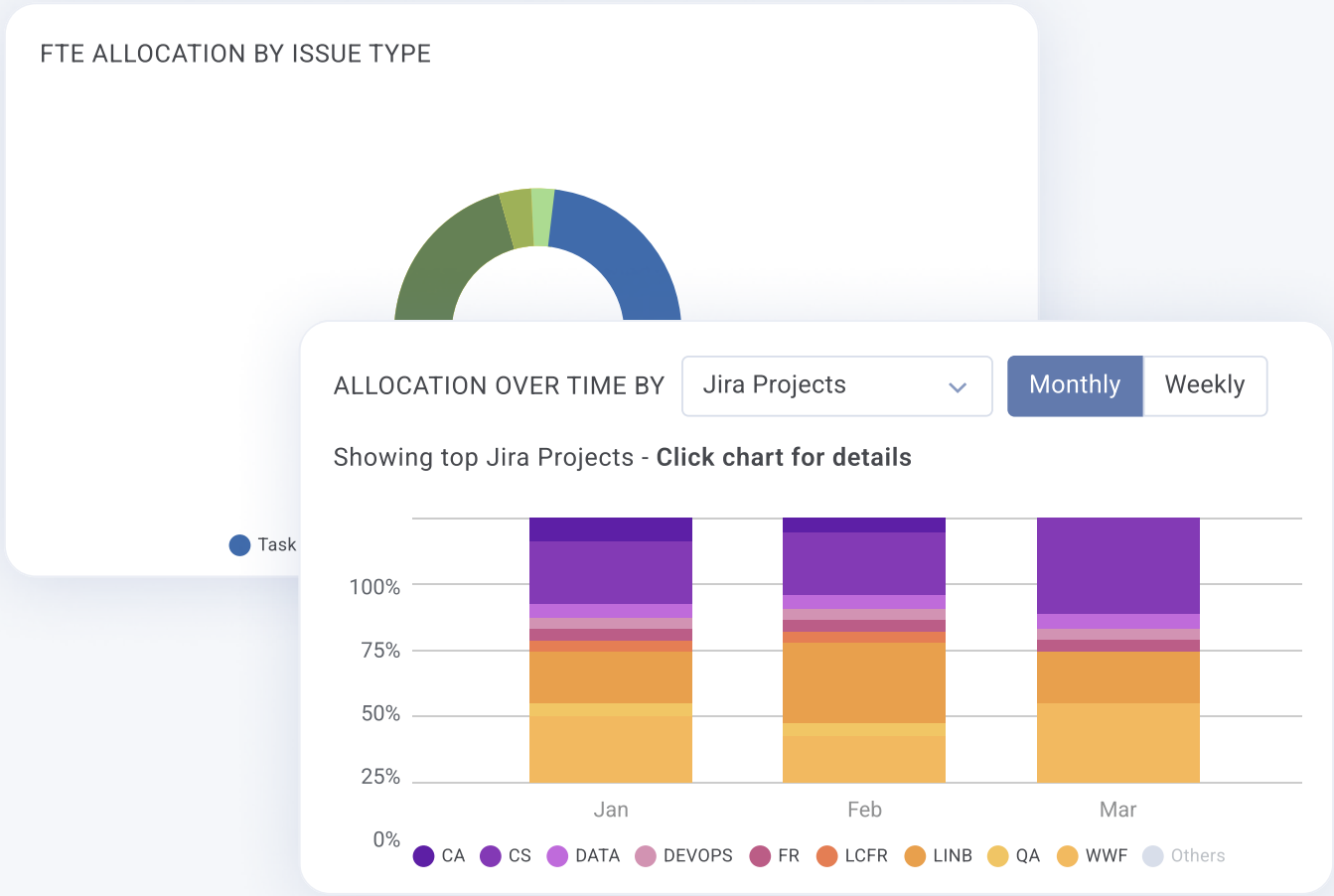
Understanding total cost of engineering initiatives



Saving money and spending wisely

Three Outcomes of Resource Allocation Visibility

Knowing what your teams are working on



A metrics program helps engineering leaders set data-backed improvement goals, set strategy, and increase operational efficiency for development teams. A metrics program that incorporates visibility into how teams are spending cycles and includes industry standard benchmarks for optimizing resourcing helps them deliver operational excellence AND drive better business outcomes.

More data means more context. More context means improved decision making.

In addition to always making the right call, resource allocation visibility also helps leaders:

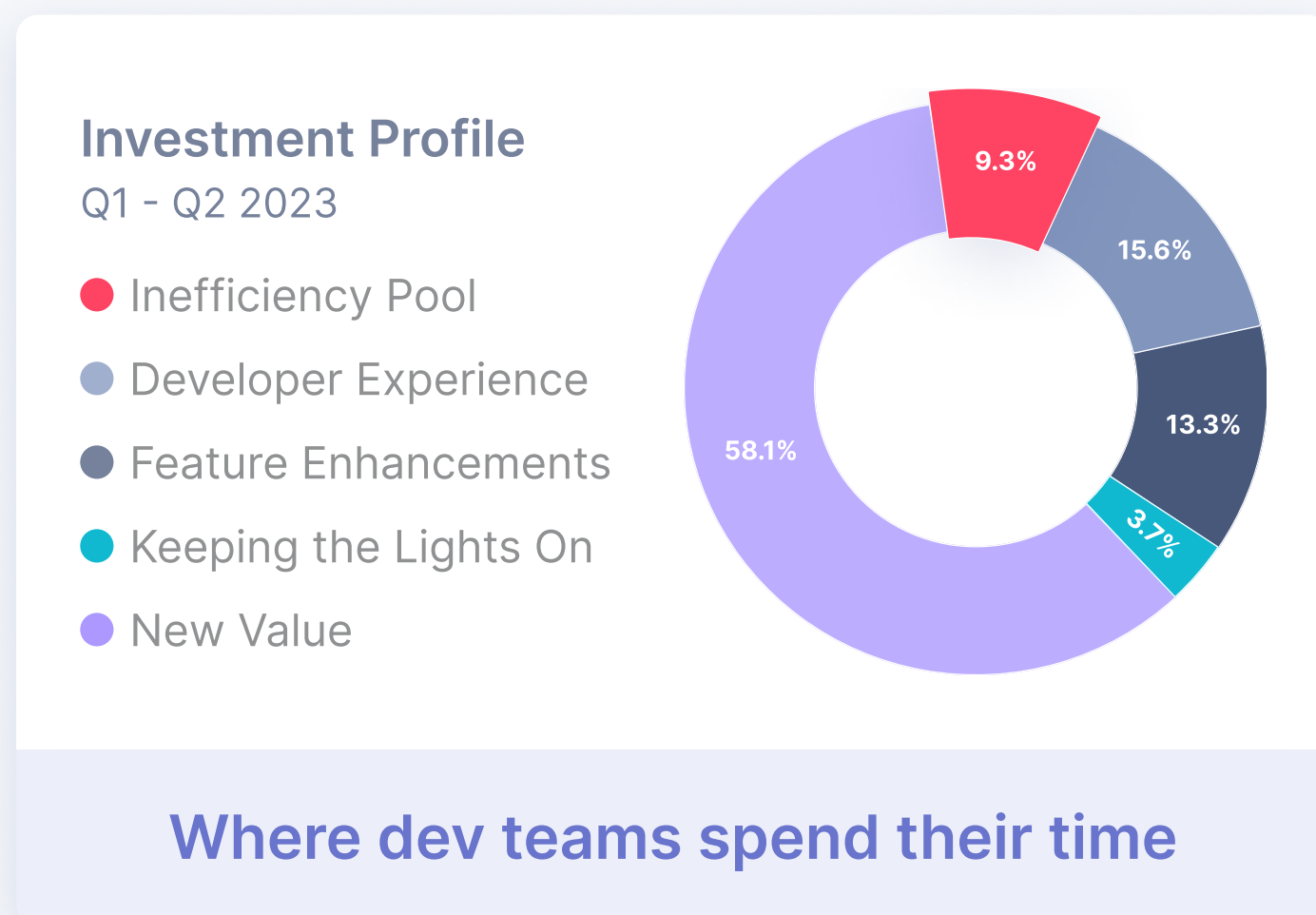
- ✓ Accurately forecast project delivery
- ✓ Flag delivery risks in order to proactively take action
- ✓ Make informed recommendations on what projects to prioritize and allocate more budget to (or reduce efforts and resources)

KEY INVESTMENTS			FTE
MW-592	Purchase experience	5.7	ON TIME
NIA-686	Infrastructure work	4.6	
EP-265	EDO M1: Admin UI GA	4.3	DELAY RISK
CM-380	Updates for Dashboard IA changes	3.6	
RN-326	Front-end UI calculation	2.4	DELAYED
CDX-1163	Profile Pages (Phase 1 - Dec 1)	2.2	

We'll deep dive into this subject in another guide, but suffice it to say that resource allocation and on time project delivery are joined at the hip. You want that high priority initiative to ship on time with high quality? You need to ensure you have enough resources working on it. Having a holistic view into how projects are resourced can highlight opportunities to reallocate headcount and budget to better align to the needs of the business. Without this visibility, engineering leaders are flying partially blind.

Three Outcomes of Resource Allocation Visibility

Understanding total cost of engineering initiatives



It's hard to know the costs or the bottom line impact of optimizing your resourcing strategy—at least until you have more visibility and context.

Just like engineering priorities, business priorities can change quickly. What was once a rallying cry for the entire company can easily get shelved—pour one out for Zune.

When business needs change, engineering must respond in kind by making resourcing recommendations (and decisions) for projects to align to a new strategic direction. And that requires a firm understanding of current and historical project resourcing. Without clear visibility into how projects are resourced—and industry insights into what peers are doing—reallocating people effort correctly can be tough.

As an example, if an engineering organization has only put 40% of their resourcing into new value (as opposed to the industry average 60%), this is a clear signal that something needs to change.

Let's assume a dev team of 100 moves some resources around and reallocates just an additional 1.5% to both (1) New Value and (2) High value projects as determined by conversations with business leaders. If we take that 3% reallocation X 100 devs X an annual cost of \$100K (a likely low number), the new value that can be reinvested is \$300K.

That's the power of visibility into engineering costs and better resource utilization. You can get a sense of how to **optimize your resourcing and find where potential value is hiding within your organization with the ROI calculator.**

Three Outcomes of Resource Allocation Visibility

Saving money and spending wisely

No company—or department within a company—is immune from economic downturns or recessions. Engineering is no exception. In the current climate of flat or shrinking budgets, every budget request for tooling or additional headcount needs to have a legitimate business purpose and the data to back it up.

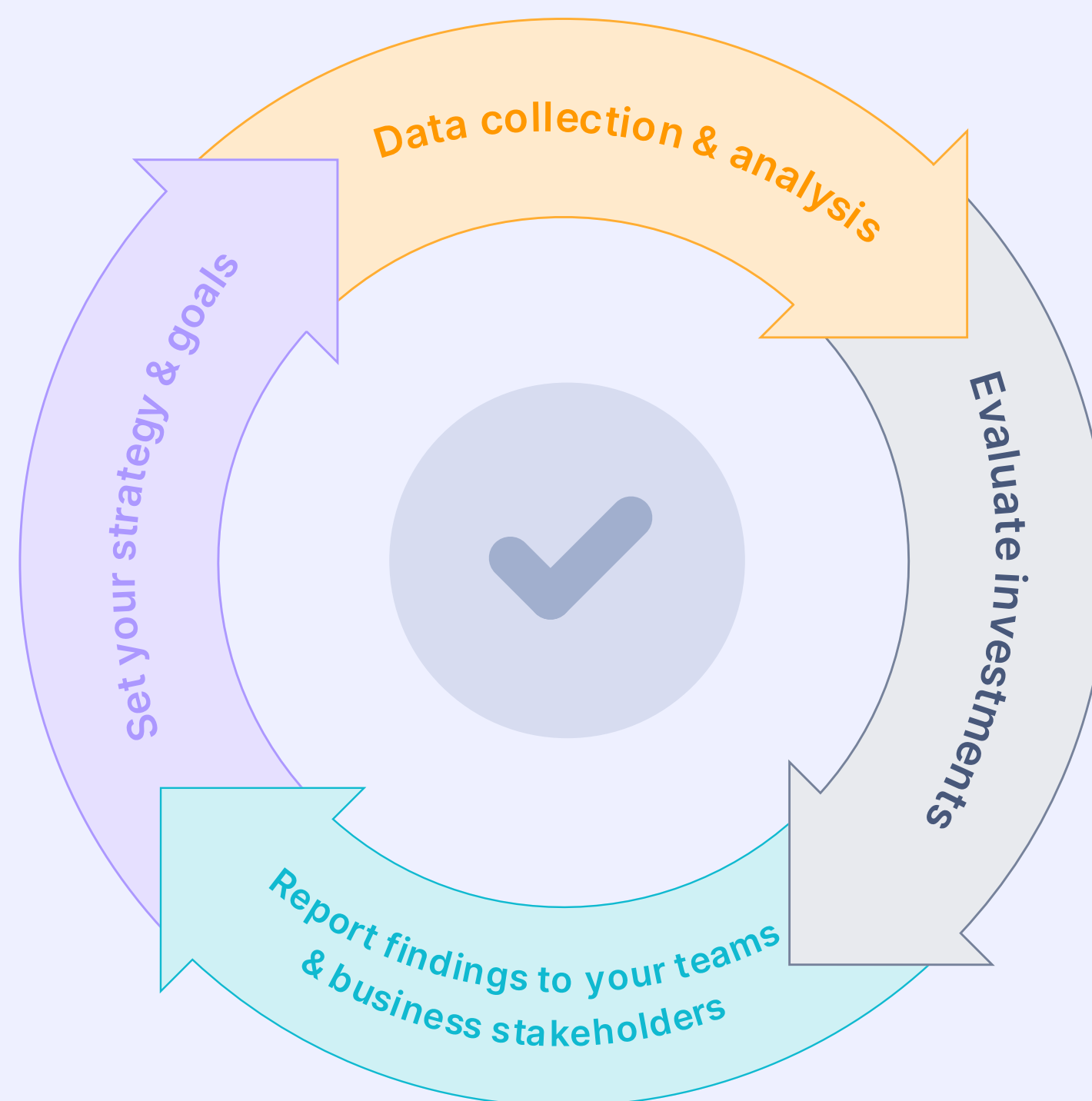
When presenting on the health, productivity, and prioritization of engineering work to the business, leaders need to include both engineering efficiency and resource allocation data. It’s only with this complete picture that productive conversations can happen and requests can be given fair consideration.

Unified report of projects, resourcing, costs, and work type

Projects Initiatives Epics Labels Area of Focus								
Type name								
Rows per page: 25 1-10 of 10								
KEY	NAME	ASSIGNEES	LAST ACTIVITY	EPICS	% OF TOTAL WORK ↓	EFFORT (FTE)	EST. COST BY FTE ⓘ	ISSUES WITH NO EPICS
LINBEE	LinearB New Dev	+41	19 Oct 2023	38	57.6%	18.1	\$ 902.6k	678
CS	Customer Support	+48	19 Oct 2023	0	28.7%	9	\$ 449.5k	437
LINB	LinearB Dev	+22	20 Sep 2023	12	8%	2.5	\$ 124.8k	146
LCFR	LinearB Customer Feature Request	+10	26 Jun 2023	0	2.8%	0.9	\$ 43.8k	68
WWF	Workflows	+5	09 Oct 2023	4	1.6%	0.5	\$ 25k	4
DATA	Data		08 Oct 2023	6	0.8%	0.3	\$ 13k	6
DEVOPS	DevOPS	ZS	04 Jun 2023	0	0.3%	0.1	\$ 3.9k	8
QA	QA	MA	29 May 2023	1	0.2%	0.1	\$ 2.6k	5

Building Resource Allocation Visibility Into Your Metrics Program

How to add resourcing to engineering improvement initiatives



In **The Engineering Leader's Guide to Building a Metrics Program** we outlined the procedure for building a metrics program from scratch.

Adding business-level reporting to a metrics program with resource allocation visibility is a similar process.

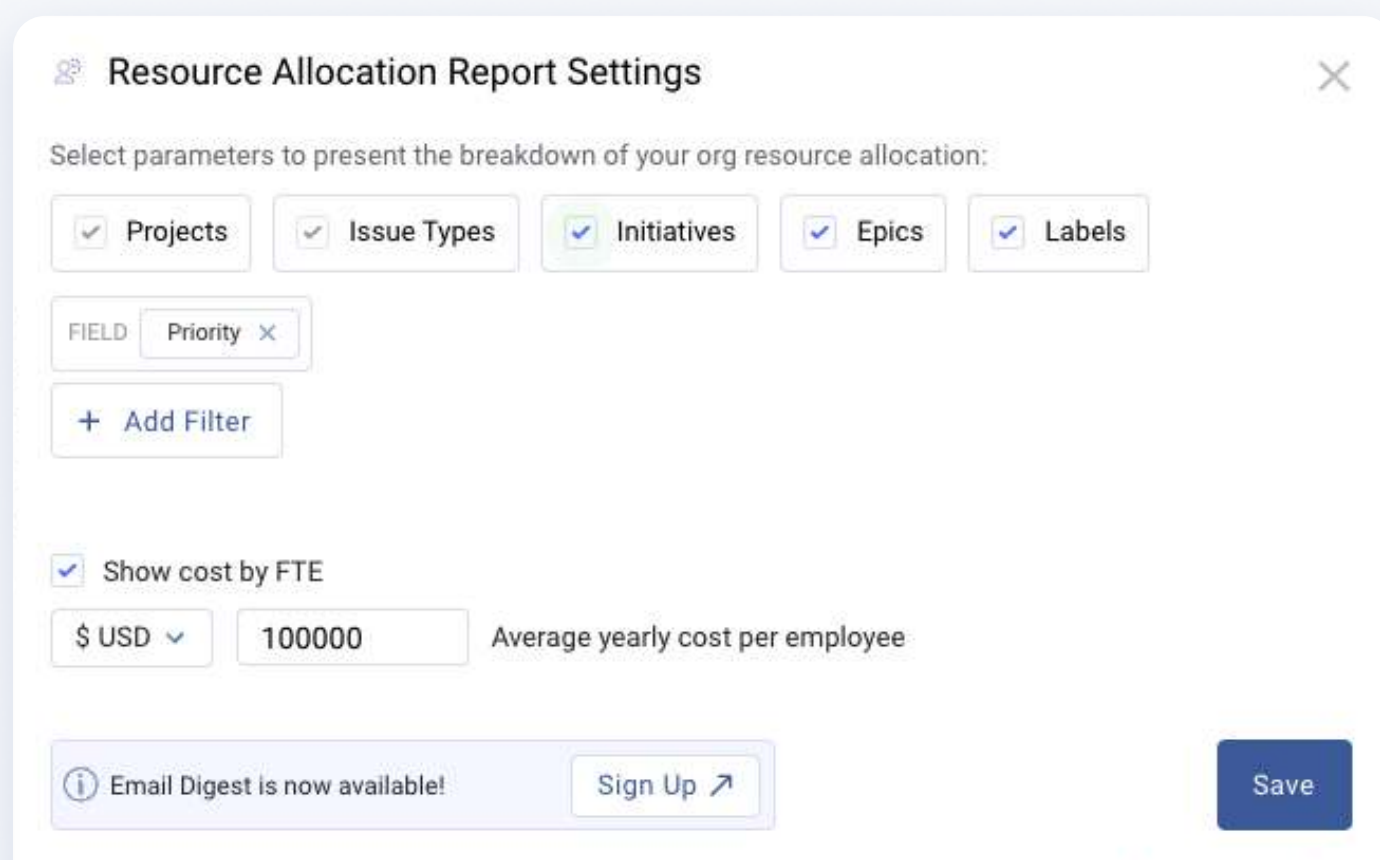
Here are the four steps to adding resource allocation to your metrics program:

- 1 Data collection & analysis
- 2 Evaluate investments
- 3 Report findings to your teams & business stakeholders
- 4 Set your strategy & Goals

Building Resource Allocation Visibility Into Your Metrics Program

STEP 1

Data collection and analysis



The screenshot shows the 'Resource Allocation Report Settings' dialog box. It includes a close button (X) in the top right corner. Below the title, it says 'Select parameters to present the breakdown of your org resource allocation:'. There are five checkboxes: 'Projects', 'Issue Types', 'Initiatives', 'Epics', and 'Labels', all of which are checked. Below these is a 'FIELD' section with a dropdown menu currently set to 'Priority' and a close button (X). There is also a '+ Add Filter' button. Further down, there is a checkbox for 'Show cost by FTE' which is checked. Below this, there is a currency dropdown set to '\$ USD' and a text input field containing '100000', followed by the text 'Average yearly cost per employee'. At the bottom left, there is a notification: 'Email Digest is now available!' with an information icon. Next to it is a 'Sign Up' button with an external link icon. On the bottom right is a 'Save' button.

Build and customize your reporting

Goal: Understand how developer time is invested across your project portfolio.

How to get there: Start by digging into your project management boards to see where all your work is being done. Make sure nothing is left behind; good reporting and decision making relies on having all the data. However your team catalogs work—by projects, initiatives, tags, labels, custom fields, issues, severity, class—account for all of it.

Did you know?

An average of 30% of work is unaccounted for in Jira—meaning there isn't a ticket or issue linked to pull requests. Correlate project data with your git systems to ensure you have visibility of all work being done.

What you're looking for: The results will be informed partially by your PM tools' structure and partially by how you organize your engineering practice though teams/projects/initiatives. You want holistic project visibility that includes FTE count, issue type investment, and project forecasting/execution metrics. Then incorporate average salary to understand Resource Allocation even better so you can **present decisions, project progress, and recommendations in terms of actual costs** that will resonate better with your business stakeholders (more on this later).

Pro Tip

Be sure to include data on the impact of planned and unplanned work. Also track and report team health indicators. This will ensure you balance work across teams and projects and avoid burnout when setting your strategy and goals.

Outcome: Holistic visibility of your projects and how they're currently resourced. Now you're ready to evaluate your current investments, report your findings, and optimize your strategy.

You could try to do all this manually, but it's extremely time consuming. Or you can automate the process of PM tool data collection and organization with LinearB ([link](#)). In addition to being no-code, LinearB is extremely flexible—showing you how things like labels and custom tags are resourced—so there's no need to change how you structure your work.

Building Resource Allocation Visibility Into Your Metrics Program

KTLO (Keeping the Lights On)

Minimum tasks a company is required to perform in order to stay operational, maintain stability, and meet stated SLAs. This might include activities such as: Maintaining current security posture, issue and ticket monitoring/troubleshooting, maintaining promised uptime, etc.

Developer Experience

Work that improves the productivity of development teams and their overall experience. Things like introducing better developer tooling, incorporating additional testing automation, restructuring code, and paying down tech debt to reduce KTLO bucket.

Feature Enhancement

Work to enhance features or deliver a product that ensures customer satisfaction. This includes activities such as: Customer RFEs, improved performance / utilization, better scalability, quality days, and improved reliability or security. etc.

New Value

Any investment in new features that increase revenue and growth by new customer acquisition or expansion. This might include activities such as: adding a new feature, supporting a new platform or partner application, implementing roadmap items, etc.

Inefficiency pool

Wasted effort due to common inefficiencies present in the SDLC—think context switching and/or idle time. This investment category is a prime candidate for reallocation of resources.

STEP 2

Evaluate investments

Goal: Prioritize your projects by high/low impact, look into current investments, and formulate initial recommendations on how to optimize resourcing. Basically translating engineering to business.

How to get there: After gathering all your data and bucketing the work into investment categories, it's time to understand what your team is currently (and SHOULD be) focused on. This is the perfect time to double check that all work is accounted for and reflected in one of the following areas: Keeping the Lights On (KTLO), Developer Experience, Feature Enhancements, and New Value.

Building Resource Allocation Visibility Into Your Metrics Program

What you’re looking for: Overall insight, patterns, trends, and improvement opportunities for current investments, project allocation, and project execution that you can report to the business. A key category to closely evaluate is your “Inefficiency pool.” Tap into this inefficiency pool by setting goals to improve **core impact metrics** (like PR size and cycle time) and by **leveraging workflow automation** to boost productivity.

Once you’ve determined where work is being done, who’s assigned to it, what the costs of all your projects are, and the breakdown of the work into investment categories, **you’ll want to review allocation benchmarks**—LinearB conducted a study on the industry and came up with a baseline of what teams should be investing in each area.

Pro Tip

Start by taking a look back at the previous quarter. For even more insight, look back on the last 6 months.

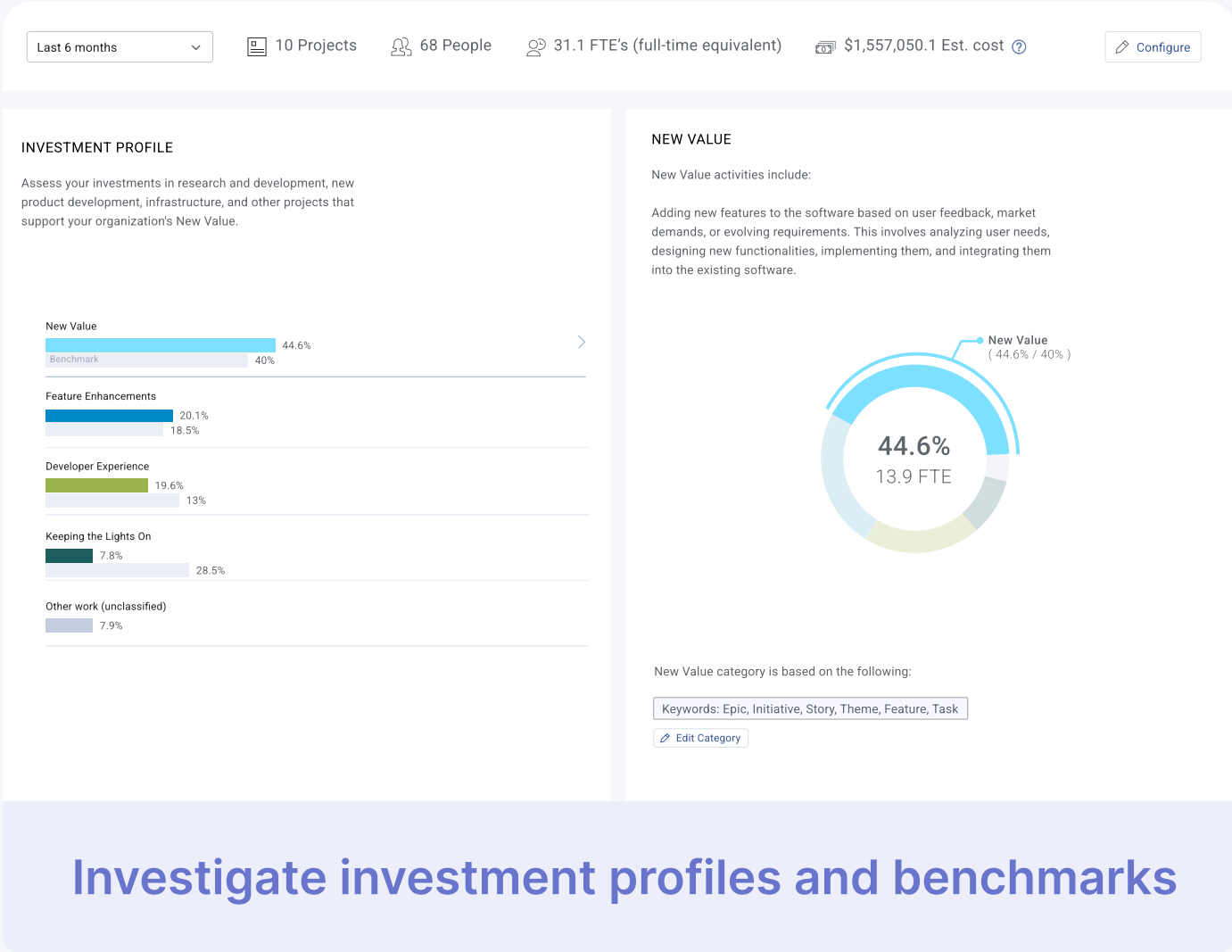
Outcome: A holistic understanding of investment profile, project resourcing, and associated costs that you can present to leadership around the business. You’ll also want to identify and evaluate low impact projects for resource reallocation. Ask yourself “Is this really something that needs X# of developers?”.

Determine high and low impact projects based on the following evaluation criteria:

- ✓ Resourcing
- ✓ Execution
- ✓ Investment categories/benchmarks
- ✓ Business priority based on stakeholders’ feedback

Pro Tip

When making project investment decisions, take your engineering expertise into consideration (not JUST business priorities). Sometimes things look good on paper but not in practice—like over indexing your investment in new value projects and not leaving enough for keeping the lights on (KTLO) work.



Investigate investment profiles and benchmarks

LinearB makes it easy to see your current project resourcing, provides guidance on best practices, and makes it easy to adjust goals to fit your specific needs. **Take a look at Investment Profile** to learn more.

Building Resource Allocation Visibility Into Your Metrics Program

STEP 3

Report findings to your teams and business stakeholders

Goal: Get everyone aligned on project allocation and present recommendations for moving forward.

How to get there: Start by identifying and reporting on the delta between investment % trend vs. expectation—begin at the organization level (which is ideal for an audience like your board of directors) then go deeper into individual team and project investment. To make these reporting sessions go smoothly, **plug your data into this template.**

Build clear reporting tailored to business audiences

Key Engineering Investments

Q1, 2023

Initiative	Business Impact	Q1 Actual		Engineering Execution			Q2 Recommend
ABC-123 Lorem ipsum dolor sit amet	HIGH	8.4	\$315K	STRONG	4d 1h	1.8	Boost
PRJ-14 Morbi id purus vitae risus sagittis	HIGH	7.8	\$293K	FAIR	6d 2h	1.1	Improve
MOB-456 Morbi id purus vitae risus sagittis	MED	7.5	\$281K	ELITE	1d 7h	2.9	Maintain
ABC-223 Integer gravida erat vitae rutrum	LOW	6.2	\$233K	NEEDS FOCUS	10d 2h	0.8	Reduce
CS-111 Sed feugiat enim elementum	HIGH	—	—	—	—	—	Start
<All other work>		7.6					

FTE | Estimated Cost (avg. cost of \$150K/dev) | Cycle Time | Merge Frequency (Merged PRs/Dev/Week)

Building Resource Allocation Visibility Into Your Metrics Program

Bringing consistent metrics to the staff meeting each week will help you transform the “mysterious” development process to a well-understood function in the business.

What you’re looking for: When reporting your findings about how project resources are allocated, try to answer common questions business stakeholders are likely to ask.

Here are some examples:

- Is feature XYZ likely to ship by the date we promised given current resourcing and operational trends?
- How quickly are we shipping new value to customers due to our current investment in new value is below the recommended 60%?
- Why aren’t we allocating more to paying down tech debt when we have so many high risk items open?

Pro Tip

Make sure to relate project investment and resource allocation to actual costs and expected ROI—especially when presenting to business stakeholders.

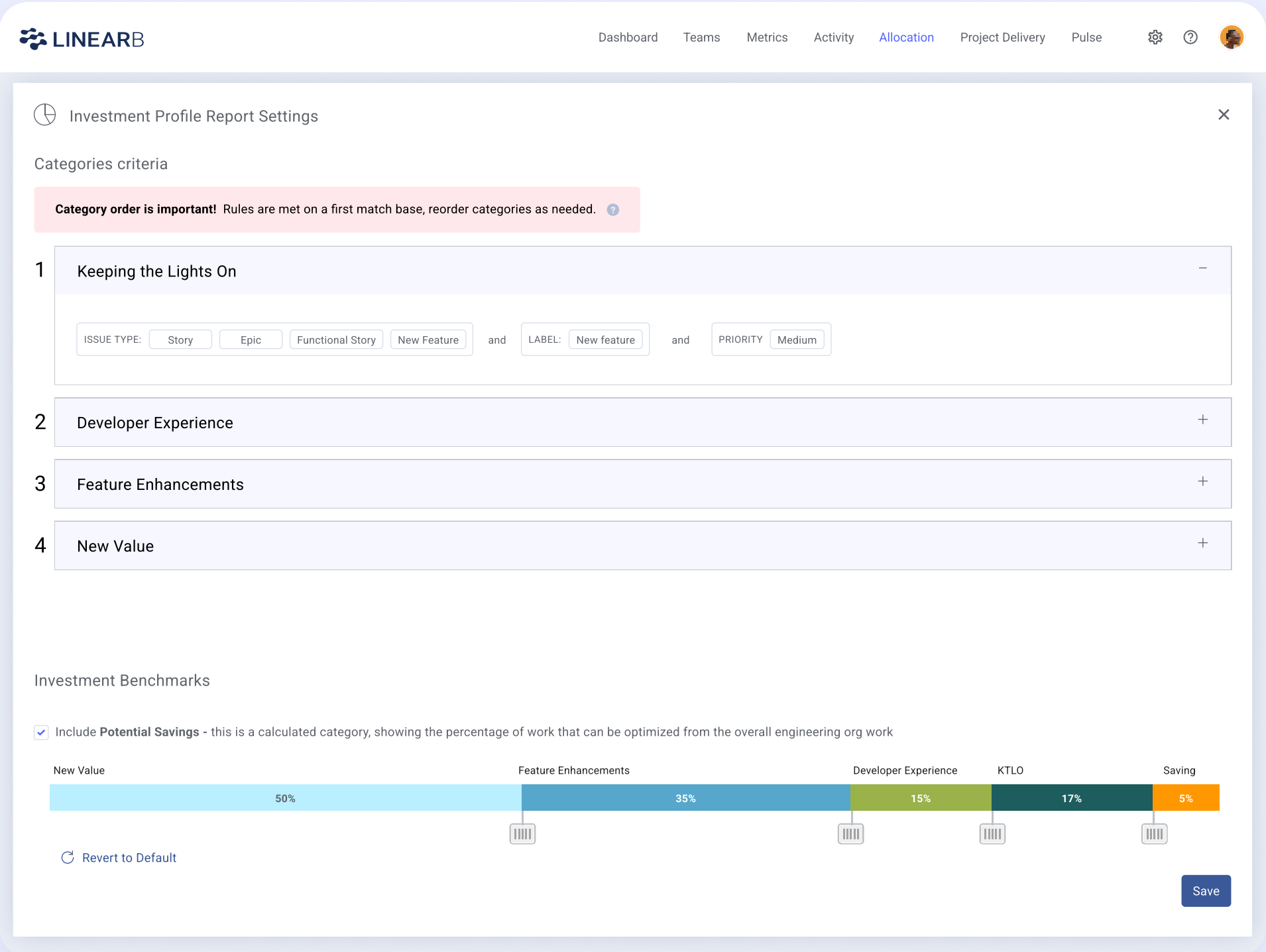
Outcome: Establish a regular cadence on allocation analysis and reporting with key stakeholders—we recommend weekly. LinearB has an automated report that goes directly to your inbox to make this process easier. Make sure to include any supporting detail for why resourcing is the way it is (like observed gaps that warrants a sprint of quality days, team health considerations, needs for better DX, etc.)

LinearB’s flexible reporting structures enables teams to segment and present their resource allocation data however it’s most useful.

Once that’s done, work with engineering and the rest of the business to figure out allocation criteria, then configure allocation benchmarks (what you should/want to invest in projects).

Building Resource Allocation Visibility Into Your Metrics Program

Customize your investment areas, benchmarks, and strategy



STEP 4 Set your strategy & goals

Goal: Set your investment benchmarks and resource allocation strategy based on current trends, identified opportunities, business needs, and stakeholder feedback.

How to get there: Use the results from your reporting session to actually put project prioritization and investment benchmarking into practice. Once you're confident in your plan for resource allocation correction, share it with your team and business stakeholders, gather final feedback on resourcing decisions and strategy, and account for any current risks not reflected in your readout.

Building Resource Allocation Visibility Into Your Metrics Program

What to look for: Use this allocation for a quarter (though change what is obviously not working) and remember it's a single data point in your overall strategy. You'll also want to look at resource allocation with additional context like:

- Project delivery and predictability impact
- Effect on operational metrics across teams/projects
- Whether your Inefficiency Pool is growing or shrinking–this is a crucial consideration for optimizing resourcing
- Cause and effect patterns like the impact of reducing DevEx investment on KTLO work
- Observed performance and trends to make reallocation recommendations for the next quarter
- Any continued work on low impact/deprioritized projects–these are opportunities to refine your allocation mix

These will be your guideposts as you optimize what the organization focuses on and how work is distributed across your teams.

Pro Tip

During the course of normal business priorities shift and needs change. If the team flags something (like not enough investment in developer experience), take that into consideration for setting and adjusting allocation and investment benchmarks on the fly.

Outcome: Once you've observed how your resource allocation mix impacts efficiency, quality, and overall engineering health, you can begin optimizing. Set strategy for next quarter along with **improvement goals (this will cascade down to operational metrics)**.

Here's an example of what that looks like in practice:

- *We will only allocate 15% of resources to keeping the lights on (bugs, tech debt, rework, refactoring). Quality will have to be high to stay under this threshold. We'll set operational team goals for: PRs under 200 diffs, Review Depth of 1, 0 PRs merged without review, <1 Risky PRs Merged per Week, Higher Deployment Frequency*

Investment Goals

Org Level OKR

Spend 15% of FTE cycles on KTLO

Team Goals

PRs under 200 diffs

0 PRs merged w/o review

<1 Risky PRs merged per week

Continue to deprioritize low impact projects/invest in business needs, focus any efficiency gains on New Value, and work to more closely **align to investment benchmarks**.

Then you'll repeat the process of observation, **reporting progress against strategy**, optimization, and goal setting for improvement.

Common Pitfalls to Avoid

Historically, engineering leaders were missing visibility into key factors such as work type investment (new value vs. KTLO), project costs, allocation across projects, and overall alignment to business needs. When working to bring resource allocation into a metrics program there are risks that you should be aware of and work to mitigate:

- Data availability and hygiene
- R&D speaks one language, the business speaks another
- Revenue visibility and engineering's distance from it

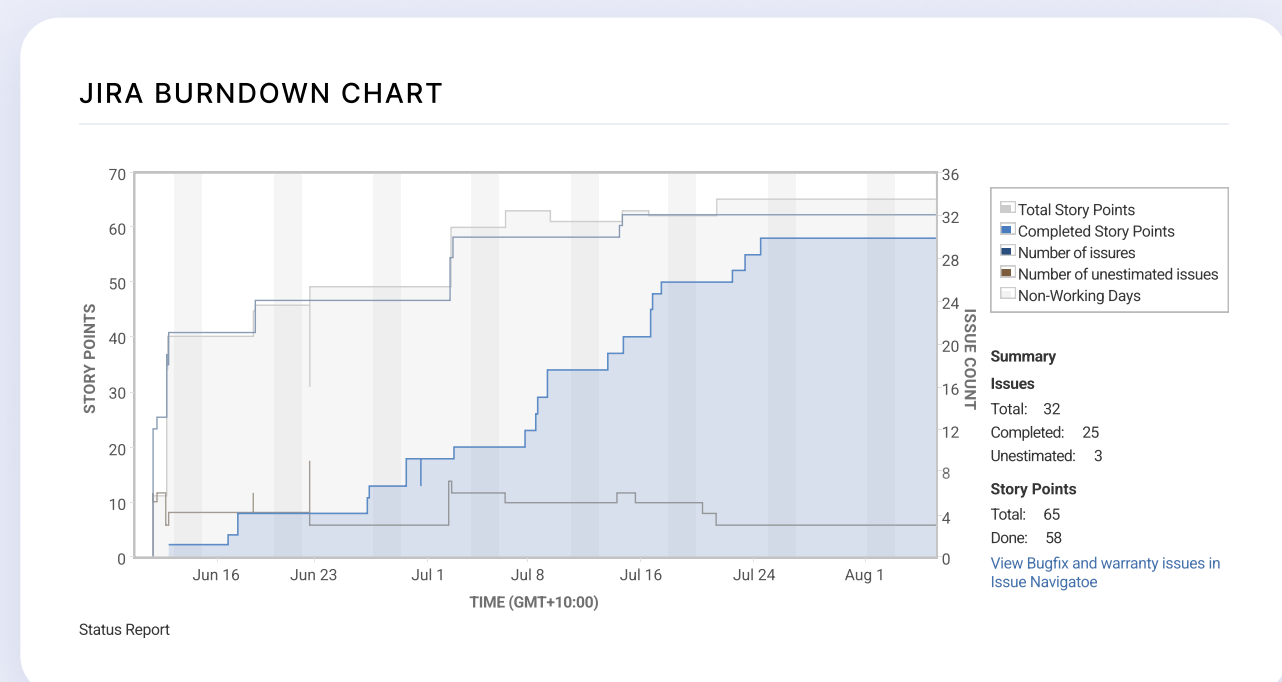
Sales leaders have Salesforce. Marketing has Marketo. But where do engineering leaders go every morning to see project health and impact on the business?

To even scratch the surface of determining whether your organization's resources are appropriated correctly and that you're focused on high impact projects requires a lot of manual effort and a significant time investment.

It's a big job to dig through PM systems, categorize the data, and finally understand the breakdown of where your teams spend their cycles (and feel confident that you're doing what's right for the business).

Data availability and hygiene

Getting visibility into resource allocation, spending, ROI, etc. has historically been a struggle because data is not always at leaders' fingertips, requires other departments (like HR), and there's always the question of whether the data can be trusted—most teams are just not great at keeping systems like Jira up to date.



This doesn't tell you much

Difficulties with translating engineering data into business outcomes

One of the hallmarks of a great engineering leader is their ability to translate engineering to executives and the rest of the business—kind of a development Rosetta Stone. Things like velocity, story points, and lines of code just don't resonate with sales or marketing teams.

Common Pitfalls to Avoid

And unfortunately these were standard metrics for health and productivity just a few short years ago—in fact some teams still use them.

Generally speaking, business stakeholders are only interested in engineering data when it can answer three questions:

- 1. Will you deliver what you promised?
- 2. Will you deliver it on time?
- 3. Are you working on things that will move the needle for the business?

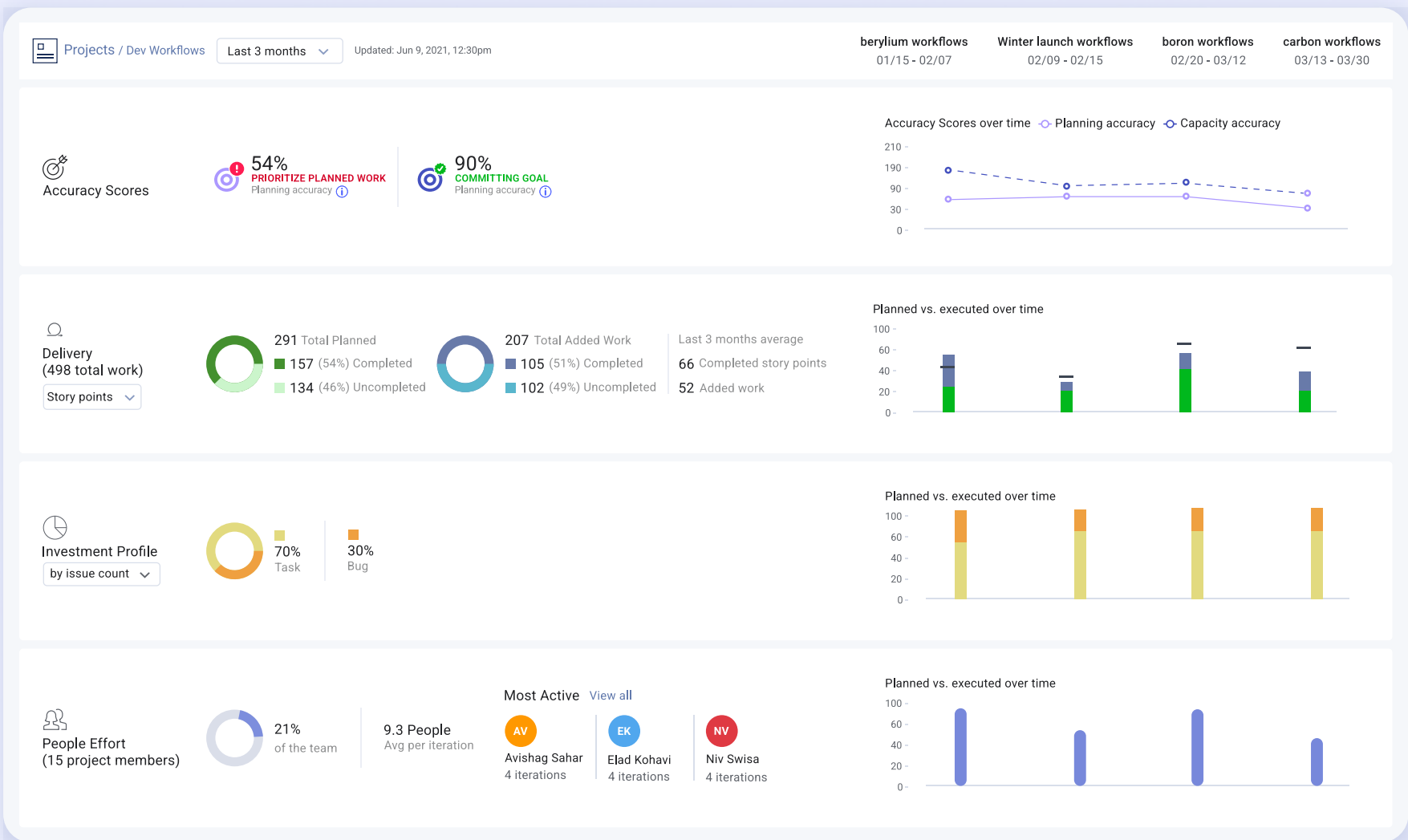
The unfortunate reality is that the “standard tools” don’t do a great job of illustrating the relationship between engineering initiatives and bottom line impact.

And while feature delivery is the shared

vocabulary that bridges the two worlds, it’s not enough.

You’ve likely spent hours listening to leaders in Sales, Marketing, Business Development, and Customer Success attempting to understand what they think would drive business. But are any of those things anything other than new features?

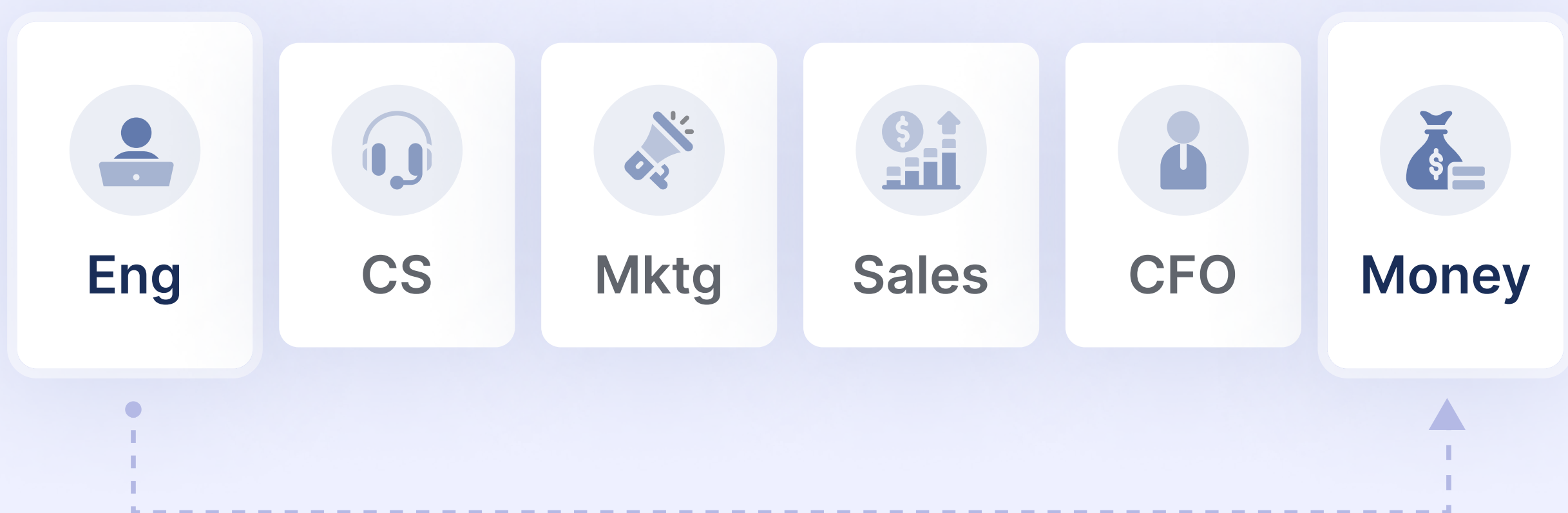
Without data, business stakeholders often can’t see the positive impact of engineering output that can’t be sold—quality days, refactoring, paying down tech debt, security enhancements, or addressing technical risks. And if they can’t see the impact of that work, they likely can’t see the downstream effect of unplanned work that was added mid-iteration.



This team is executing well, but unplanned work is bogging them down

Common Pitfalls to Avoid

Engineering is the **furthest away**
from revenue visibility



Because there isn't connective tissue between engineering work, costs of projects/initiatives/epics/etc., and the bottom line, engineering has struggled to provide quantitative ROI, showcase alignment to business priorities, and make the case for additional headcount/budget with data.

There just isn't visibility (or at least not easily produced and understood visibility) into what engineering is working on, how

their projects are resourced/prioritized, and what "should" be resourced and prioritized.

Frequent check-ins, data-backed resourcing strategy sessions, and regular reporting on project progress (and identified risks) with business leaders are good ways to avoid these hazards and ensure that your engineering organization is tightly aligned to the business and contributing to its health.

Conclusion

LinearB's software delivery management platform is ideal for building an engineering metrics program from scratch and aligning engineering to the needs of the business. After onboarding and initial configuration/integration with your existing software stack, you'll get the metrics, dashboards, and insights discussed in this guide right out of the box.

You can **get your DORA metrics with a free forever DORA Dashboard** and begin building your engineering metrics program today! If you'd like to discuss any of what was covered in this guide in more detail, or you want to see Resource Allocation and Investment Profile in action, you can **schedule a demo**.

We'd love to hear from you!